

Qatar-2b

R-1882

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_140421 · created 2026-08-03T11:55:23+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456768.838887 +/- 9e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.21 +/- 0.039
Transit depth (Rp/Rs)^2	4.4 +/- 1.63 [%]
Orbital Inclination (inc)	85.64 +/- 1.75
Airmass coefficient 1 (a1)	1.033 +/- 0.033
Airmass coefficient 2 (a2)	-0.016 +/- 0.05
Scatter in the residuals of the lightcurve fit is	3.26 %
Best Comparison Star	#1 - [593, 96]
Optimal Method	PSF photometry
Transit Duration (day)	0.0718 +/- 0.0078

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.21 ± 0.039 vs 0.16327 (deviation 1.16σ)
Duration fit vs published	0.0718 d vs 0.0767 d (deviation 0.6σ)
Mid-time O-C	-0.0 min
Depth SNR	5.2
Residual scatter	3.26 %

Light curve

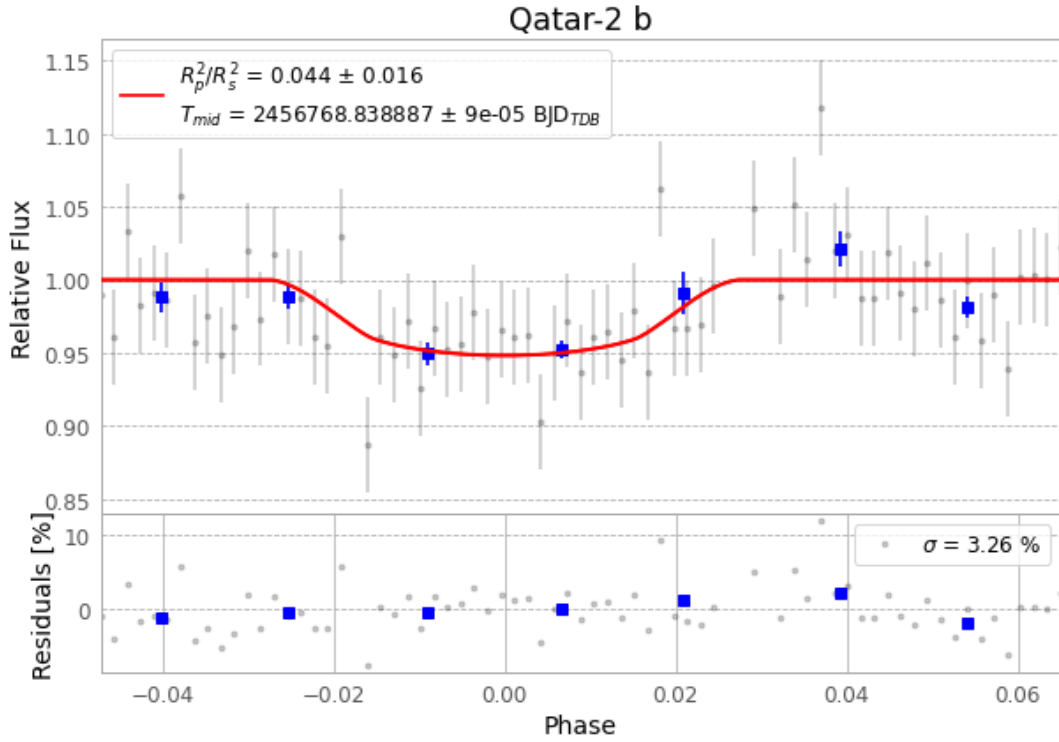


Figure 1 — FinalLightCurve_Qatar-2 b_2014-04-20.png (SHA-256 11fe50ec7f35f76d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [268, 237], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fc6eed76c96c468...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), Qatar - 2140421062712.FITS → Qatar - 2140421100312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-140421.log (a39b0185f342...), FinalLightCurve_Qatar-2 b_2014-04-20.png (11fe50ec7f35...), FinalLightCurve_Qatar-2 b_2014-04-20.csv (72660628c182...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 11:55Z.